



Problem

the test cases without optimizing to eliminate the **overlapping subproblems**. Think of a way to store and reference previously computed solutions to avoid solving the same subproblem multiple times. * Consider the degenerate cases:

- How many ways can you make change for **0** cents? - How many ways can you make change for **> 0** cents if you have no coins? * If you're having trouble defining your solutions store, then think about it in terms of the base case (**$n = 0$**).
- The answer may be larger than a **32**-bit integer.

Submissions

Sample Input 0

```
4 3
1 2 3
```

Sample Output 0

```
4
```

Explanation 0

There are four ways to make change for **$n = 4$** using coins with values given by **$C = [1, 2, 3]$** :

1. **$\{1, 1, 1, 1\}$**
2. **$\{1, 1, 2\}$**
3. **$\{2, 2\}$**
4. **$\{1, 3\}$**

Sample Input 1

```
10 4
2 5 3 6
```

Sample Output 1

```
5
```

Explanation 1

There are five ways to make change for **$n = 10$** units using coins with values given by **$C = [2, 5, 3, 6]$** :

1. **$\{2, 2, 2, 2, 2\}$**
2. **$\{2, 2, 3, 3\}$**
3. **$\{2, 2, 6\}$**
4. **$\{2, 3, 5\}$**
5. **$\{5, 5\}$**

Leaderboard

Discussions

[Change Theme](#)Language Java 15

```
1  import java.util.*;
2
3  class Solution {
4      public static long getWays(int n, List<Long> coins) {
5          long[] dp = new long[n + 1];
6          dp[0] = 1;
7          for (long coin : coins) {
8              for (int amount = (int) coin; amount <= n; amount++) {
9                  dp[amount] += dp[amount - (int) coin];
10             }
11         }
12         return dp[n];
13     }
14     public static void main(String[] args) {
15         Scanner sc = new Scanner(System.in);
16         int n = sc.nextInt();
17         int m = sc.nextInt();
18         List<Long> coins = new ArrayList<>();
19         for (int i = 0; i < m; i++) {
20             coins.add(sc.nextLong());
21         }
22         System.out.println(getWays(n, coins));
23     }
24 }
```

Line: 3 Col: 17



Upload Code as File

Run Code

Submit Code



Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0



Test case 1



Test case 2



Test case 3



Test case 4

Compiler Message

Success